

DoangJoo "Alan" Synn

Ph.D. Student in Computer Science · Georgia Institute of Technology

Advised by Prof. HyunJoo Oh and Prof. Sehoon Ha

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Research Interests

I study **motion** as a computational medium across HCI and computer graphics. My research builds systems for designing, generating, and fabricating motion. I aim to make motion programmable and tangible across animated characters, mechanical artifacts, robotic systems, and AI-assisted creative tools.

Before starting my Ph.D., I spent eight years building software systems across AI systems, games, cloud platforms, and social robots.

Education

Georgia Tech, Ph.D. in Computer Science

2022–Present

- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/ Graphics)

Korea University, M.Eng. in ECE

2020–2022

- Thesis: *On Efficient Data Delivery for Machine Learning Systems*
- Advised by Prof. Jong-Kook Kim

Korea University, B.Eng. in EE

2012–2020

- Alternative military service as a Software Engineer, 2016–2019

Publications

Refereed Conference Papers

- 1 **D. Synn**, Z. Guo, S. Ha, and H. Oh, “MotionSmith: A Sketch-Based Design System for Automata Making,” *CHI*, Apr. 2026.
- 2 J. Yu, P. Garg, **D. Synn**, and H. Oh, “Tangible-MakeCode: Bridging Physical Coding Blocks with a Web-Based Programming Interface for Collaborative and Extensible Learning,” *CHI*, Apr. 2025.
- 3 **D. Synn** and J. Kim, “Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders,” *ACK*, Oct. 2021.
- 4 J. Park, **D. Synn**, and J. Kim, “A Survey on the Advancement of Virtualization Technology,” *KIPS*, May 2021.
- 5 B. Ahn, **D. Synn**, M. Derkani, E. Ebrahimi, and H. Esmaeilzadeh, “Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection,” *NeurIPS Demo*, 2021.
- 6 D. Kim, **D. Synn**, and J. Kim, “SG-Drop: Faster Skip-Gram by Dropping Context Words,” *KIPS*, Nov. 2020.

Refereed Journal Articles

- 1 M. Ryu, X. Piao, J. Park, **D. Synn**, and J. Kim, “ADaPT: An Automated Dataloader Parameter Tuning Framework using AVL Tree-based Search Algorithms,” *TKIPS*, Jan. 2025.
- 2 **D. Synn**, X. Piao, J. Park, and J. Kim, “Distributed Training of Deep Learning Models Using Micro Batch Streaming,” *KTSDE*, Mar. 2023.

- 3 X. Piao, **D. Synn**, J. Park, and J. Kim, "Enabling large batch size training for dnn models beyond the memory limit while maintaining performance," *IEEE Access*, 2023.
- 4 J. Han, **D. Synn**, T. Kim, H. Jung, and J. Kim, "Feature Based Sampling: A Fast and Robust Sampling Method for Tasks Using 3D Point Cloud," *IEEE Access*, Apr. 2022.

Preprints

- 1 **D. Synn**, S. Ha, and H. Oh, "MotionSmith for Young Makers: Educator Co-Design and Enactment of a Mechanical CAD System," *In review*, 2027.
- 2 J. Park, **D. Synn**, X. Piao, and J. Kim, "Dataloader Parameter Tuner: An Automated Dataloader Parameter Tuner for Deep Learning Models," *arXiv*, 2022.

Professional Experience

Ph.D. Research Intern, **Dolby Laboratories**

2024–2025

- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, **LG Uplus Corp.**

2023

- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, **Protopia AI Inc.**

2021–2022

- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.
- Led framework development and Kubernetes/GPU infrastructure integration; the work contributed to a NeurIPS 2021 Demo.

Cloud Architect · **AWS Educate Cloud Ambassador**

2020–2022

- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.
- Built cloud-native infrastructure using Kubernetes, Terraform, CI/CD, observability, and cost-optimized autoscaling while mentoring early-career engineers.

Server Engineer, **Flysher Inc.**

2019

- Shipped backend features for four social-game titles; one reached Facebook's global top 10.
- Built an ETL-to-Redshift analytics pipeline supporting data-informed feature rollout.

System Engineer, **AKA Intelligence (Musio)**

2017–2019

- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.
- Coordinated hardware-software integration and production QA with overseas ODMs.

Teaching

Head TA, CS 4496/7496 Computer Animation

2024–Present

- **Georgia Tech, School of Interactive Computing.**

Lecturer, Korea University*2021–2022*

- Taught introductory Java programming to 10+ undergraduate students.

Teaching Assistant, Korea University*2021*

- Digital System Design
- Data Structures and Algorithms

Lecturer, KAIST Engineering School*2012–2014*

- Co-developed national project-based coding education programs with South Korea's Ministry of Science and ICT.
- Designed and evaluated 20+ curricula combining Arduino, hardware prototyping, and software development for high-school and college students.

Invited Talks & Outreach

Invited Panelist and MotionSmith Demonstrator, AutomataCon*2026***Motion, Making & Mechanical Design — GIFT Workshop, Georgia Tech CEISMC***2026***Invited Talk, Korea University***2026***MotionSmith Presentation, CHI***2026***Honors & Awards**

Fellowships

- [Brain Korea 21 \(BK21\)](#) Research Fellowship (2020–2022)

Awards

- Prize Winner, Creative Challenge Program @ [Korea Univ. CTL](#) (2015 and 2016)

Activities & Service

Georgia Tech Korean Student Association*2023–2024*

- President of Korean Students in the College of Computing.

Hardware & Software Community, Korea University*2014–2019*

- President; grew the community from 10 to 200+ active members across hardware and software tracks.

TEDxKoreaUniversity, Korea University*2015–2016*

- Founder & Chief Organizer; led a 47-person team in producing a 200-guest event with 7 speakers and a \$4K budget.